

S&P 500 Stock Market Analytics

TECHNICAL DOCUMENTATION & PIPELINE PROJECT REPORT

This technical document details the end-to-end data pipeline implementation for the S&P 500 Stock Market Analytics system. The report outlines the complete process, including initial data preparation, data cleaning operations, backend SQL logic formulations, front-end metrics calculations, and time-series visualization configurations.

1. Data Extraction & Cleaning Operations

Raw financial stock records often contain format inconsistencies and processing discrepancies. The following systematic pipeline stages were implemented to establish data integrity:

- **Handling Empty and Null Rows:** Missing records or rows containing empty stock quotes were dropped from the operational dataset. This filtering ensures that metrics such as historical close averages and volumes are calculated without distortions.
- **Data Type Transformations & Conversions:** Field parameters were adjusted to their ideal computational data types:
 - The date string format was successfully converted into the standard SQL YYYY-MM-DD date format.
 - The volume field was cast as BIGINT to prevent mathematical overflow errors, given that aggregate index volumes routinely scale into billions and trillions.
 - Pricing fields (open, high, low, close) were mapped into fixed-decimal formats, specifically DECIMAL(10,2), for precise fractional values.
- **Deduplication:** Repeated ticker entries recorded at identical processing timestamps were eliminated using programmatic row-deduplication logic.

2. Backend Analytics & Key SQL Logic

Once structured database schemas were applied, target SQL script blocks and dynamic computational expressions were created to capture high-impact market parameters.

A. Filtering the Top 5 Traded Symbols by Volume

To pinpoint high-liquidity stocks with major market turnover, the following grouping and ordering query was executed:

```
SELECT
    symbol,
    SUM(volume) AS Total_Trading_Volume,
    AVG(close) AS Average_Closing_Price
```

```
FROM public_market_table
GROUP BY symbol
ORDER BY Total_Trading_Volume DESC
LIMIT 5;
```

This backend segment identifies companies experiencing significant market interest and provides insights into underlying investment trends.

B. Dynamic Price Volatility Index Calculation

To isolate high-risk swing trading opportunities, a dedicated volatility metrics measure was created using a standard range spread calculation:

```
Price_Volatility = AVERAGE(public_market_table[high]) -
AVERAGE(public_market_table[low])
```

In the workspace layout, this expression powers a ranked visualization block, displaying the absolute top 5 volatile tickers based on any active search query configurations.

3. Visualization Mapping & Trading Logic

The time-series workspace was split into two primary analytic tools to provide detailed operational perspectives over explicit observation intervals:

A. Primary Closing Price Trend Framework

- **X-Axis Configuration:** Linked to the baseline date attribute. Daily data fluctuations were filtered out by bypassing minor granular levels (Quarters, Months, Days) to build a clean, structured, ****Year-over-Year trend timeline****.
- **Y-Axis Specification:** Map tracking is linked to the close variable. The consolidation mode is explicitly set to Average of close to cleanly reveal historical price movements.

B. Algorithmic Moving Average Crossover Mechanics

To reduce short-term price variations and expose valid trend changes, dual rolling average scripts were formulated:

- **7-Day Moving Average (Fast Trend Line):** Captures immediate short-term momentum and price breakouts.
- **30-Day Moving Average (Slow Trend Line):** Serves as a macro baseline to reveal broader underlying trends.

```
7_Day_MA =
AVERAGEX(
    DATESINPERIOD(public_market_table[date], LASTDATE(public_market_table[date]), -7,
    DAY),
```

```
CALCULATE(AVERAGE(public_market_table[close]))
)
```

```
30_Day_MA =
AVERAGEX(
    DATESINPERIOD(public_market_table[date], LASTDATE(public_market_table[date]), -30,
DAY),
    CALCULATE(AVERAGE(public_market_table[close]))
)
```

4. Analytical Conclusions & Trading Interpretation

The operational logic rules governing data interactions across the final interface dashboard are summarized below:

Dashboard Module	Underlying Metric System	Strategic Core Function
Top 5 Traded Companies	Aggregated Volume: SUM(volume)	Highlights highly liquid entries with narrow transaction slippage.
Top 5 Volatile Companies	Absolute Price Spread Range	Identifies active price-swing setups for short-term trading.
Golden Crossover (Buy Alert)	7_Day_MA > 30_Day_MA Intersection	Signals an upward price shift, highlighting key asset accumulation windows.
Death Crossover (Sell Alert)	7_Day_MA < 30_Day_MA Intersection	Signals downward market momentum, highlighting potential risk mitigation or exit points.

User Interface Feature: The dynamic filter panel on the left side features an embedded text-search option. This lets users instantly isolate any specific ticker among 500+ assets, automatically refreshing all connected charts, averages, and KPIs.